

	1.8 × 10 ⁵ N	
3(a)	▪ The gas consists of a <i>very large number of molecules in continuous random motion</i>. ▪ There are <i>no intermolecular forces</i> between molecules except during collisions between molecules. ▪ The gas molecules undergo <i>elastic collisions</i> with one another and with the walls of the container. ▪ The <i>duration of collisions between molecules or between molecules and the walls of the container is negligible</i> compared to the time interval between collisions. ▪ The <i>volume of the gas molecules is negligible</i> compared to the volume of the container.	
3(b)(i)	Using $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$. $\Rightarrow p = \frac{1}{3} \rho \langle c^2 \rangle$ $\Rightarrow \sqrt{\langle c^2 \rangle} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3(1.8 \times 10^5)}{2.7}}$ $= 447 \text{ m s}^{-1}$ $= 450 \text{ m s}^{-1}$	
3(b)(ii)1.	Internal energy of an ideal monatomic gas is given by $U = \frac{3}{2} nRT$ Since volume is doubled at constant pressure, its temperature is doubled. $\Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} (2.5)(8.31)(576 - 288)$ $= 8.97 \times 10^3 \text{ J}$	
3(b)(ii)2.	Work done $W = p\Delta V = nR\Delta T$ $= (2.5)(8.31)(576 - 288)$ $= 5.98 \times 10^3 \text{ J}$ Since the volume of the gas increased, work is done by the gas. Thus, work done on the gas = $-5.98 \times 10^3 \text{ J}$	
3(b)(ii)3.	Using 1st law of thermodynamics, $\Delta U = Q + W$ $Q = \Delta U - W$ $= 8.97 \times 10^3 - (-5.98 \times 10^3)$ $= 1.49 \times 10^4 \text{ J}$	

4(a)	The principle of superposition states that when two or more	
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